Knotted Coil Designer

A browser tool for designing MRI receive phased arrays made of **knotted wire loops** (torus knots T(p,q), with 1–8 lobes) on a helmet former. It generates single elements, “blooms” them, and **automatically arranges N elements to reduce coupling between neighbours (geometric decoupling)**. It exports wire paths, coupling matrices and meshes.

> **Status: exploratory research software.** Every electromagnetic output is a *quasi-static geometric estimate*: it assumes low frequency and models the wires as thin current paths. It ignores capacitors, the head/body load and wave effects. Confirm any design with a full electromagnetic (full-wave) simulation or with bench S21, the coupling between two coils measured on a network analyser, before building.

How to run

- **Easiest:** double-click `app/index.html`. It needs no install and no internet; the Three.js 3D library is stored in `vendor/`.
  - *Not yet tested:* opening by double-click. It was tested served from a local web server.
  - If the array solver doesn’t respond when opened by double-click, use the built-in local server (tested). Run `node tools/serve.js` in this folder, then open http://localhost:8765/app/index.html.
- **Validation suite** (needs Node.js; takes about 5–15 minutes):

```
node tests/run.js
```

This writes `validation/validation_report.html`. The app’s header shows the latest verdicts.

What’s where

Path	Contents
`DEVELOPER_GUIDE.md`	**Start here if you want to extend the tool:** architecture, how to add knot families, rules
`CLAUDE.md`	Instructions for AI coding assistants working on this project
`docs/REQUEST_LOG.md`	The owner’s design requests, quoted, and what each produced
`docs/research_integrity/`	The research-integrity rules this project follows (Reza Farivar’s framework, DRAFT v1.0)
`sources/`	Archived primary sources for verified citations
`NOTICE.txt`	Copyright, sharing terms, status
`Design.md`	Frozen design v1.10 (v1.8 = knot thickness T; v1.9 = crossover bumps between overlapping coils; v1.10 = COMSOL interface). The changelog records every version; checks V1–V25 were each pre-registered before their code ran
`DESIGN_HISTORY.md`	One-page history of every version, trigger, check and outcome
`sim/`	Full-wave capacitor co-simulation (Python + openEMS), v1.7; pre-run geometry gate `sim/check_crossings.py` (v1.9)
`docs/AUDIT_*.md`	The night-of-2026-10-20 audit: brainstorm, independent review, four area audits
`DECISIONS.md`	Dated decision log, including validation failures and open questions for Reza
`PARAMETERS.md`	Every parameter with its origin tag (LIT / CONV / JUDG / TUNED / DERIVED)
`core/`	Geometry and EM package: `knot.js`, `inductance.js`, `surface.js`, `layout.js`, `optimise.js`
`app/index.html`	Browser UI: Element, Pair explorer, Array on helmet
`tests/run.js`	Pre-registered validation checks plus labelled post-hoc checks
`tests/compare_worstpair.js`	Pre-registered solver comparison
`tools/serve.js`, `tools/build_dist.js`	Local server; builds `dist/` (a flat, hostable copy of the app)
`validation/`	Generated validation reports
`vendor/`	Three.js 0.149.0 (MIT licence, `vendor/three-LICENSE.txt`)

Validation status (full run 2026-10-23 11:48, code d3cfeb9 plus the uncommitted changes of DECISIONS 2026-10-23, Design v1.10)

The report records git d3cfeb9 because the 2026-10-23 changes were not yet committed when it ran. Every verdict is identical to the run of 2026-10-21 (code bc374d0).

Pre-registered checks, applied mechanically. Full report: `validation/validation_report.html`.

Check	Verdict
V1 Maxwell coaxial mutual inductance	PASS (worst error 0.005%)
V2 Loop self-inductance	PASS (worst error 0.66%)
V3 Coplanar-circle null spacing	NOT EVALUABLE (reference unverified; computed 0.760 × diameter)
V4 Convergence · V5 Symmetry	PASS
V6 Topology across bloom (β 0.02–3.0)	**FAIL** only below the wire-dependent bloom floor (β = 0.02, 0.05), a physical limit shown in the app
V7 Topology across petal sharpness	PASS
V8 B1 on-axis loop field	PASS (0.002%)
V9 Ring-layout symmetry (pre-bump layout) · V10 Capacitor breaks · V11 Capacitor value · V12 Face zones · V13 Reproducibility	PASS
V14 B1 vs Magpylib (independent library)	PASS (< 0.0001%)
V15 Full-wave loop resonance with first-pass capacitors	**FAIL** (−10.3%; thin-wire effective radius)
V15b Thin-wire correction carries over (trefoil)	**FAIL** (+6.7% vs 5% limit)
V16 S-matrices passive and reciprocal	**FAIL** on loop_air_3T (1.0011) and trefoil_air_7T (1.0013); other cases pass
V17 Tuning and matching	PASS
V18 Quick B1 map vs full-wave near the surface (3T, 7T)	**FAIL** at both (typical point 7–24% off; worst points at near-zero field)
V19 Pair null survives capacitors and load	NOT RUN (pair simulations in progress; the first attempt was invalid, see DECISIONS.md)
V20 Second solver (mariepy)	NOT EVALUABLE (not run)
V21 Thickness control · V22 Thick-knot B1 vs Magpylib · V23 Tilted-circle self-inductance	PASS
V24 Crossover bumps	**FAIL**: all 4 pair cases pass; all 4 test arrays fail (dense knot arrays cannot all be separated)
V25 COMSOL vs openEMS	NOT EVALUABLE (COMSOL not installed on the development machine)

Ring vs free layouts (C2, pre-registered): the ring layout met none of its 3 criteria, so free layouts stay the default.

Every failing run is kept in the git history; see DECISIONS.md.

Proposed v1.11 (2026-10-23, pending Reza's approval) — COMSOL link in the app, EXPLORATORY:

- The Array tab's "COMSOL" card and the Pair explorer's "Verify in COMSOL" send the design to COMSOL and run it in the background. You need `node tools/serve.js`, a licensed COMSOL with the RF Module and a Python with MPh; set-up for Windows and Linux is in DEVELOPER_GUIDE "v1.11".
- One or two coils are tuned and matched inside COMSOL. Arrays get one port per coil (no capacitors, no tuning or matching) and give B1 maps and coupling only; their single-coil maps are distorted by the missing capacitors, the whole-array map is the fair comparison.
- Every case is checked before it is sent (coil too large for the head, wires inside the head or touching, gaps next to another coil).
- The app then shows COMSOL's capacitor values, S-parameters and coupling next to its own numbers, $B1^-$ maps side by side with a ratio map, and COMSOL points on the Pair explorer graph.
- Insulation beads (placeholder permittivity) sit at every crossing (one or two coils).
- `python sim/comsol_run.py case.json` does the same without the app. "Download case + STL" writes the case file.
- No pass/fail criteria (requester's choice). Tested on Linux with COMSOL 6.4; written for Windows but not yet run there.

New in v1.10 (COMSOL interface, 2026-10-21) — UNVALIDATED:

- `python sim/export_comsol.py sim/cases/<case>.json` writes plain files for COMSOL into `sim/comsol/<case>/`: wire segments between capacitor gaps (x y z, mm), a ports table, the head phantom and metadata. It needs numpy only.
- `python sim/comsol_build_mph.py <case>` builds, solves and extracts the model through MPh (Python ↔ COMSOL). It needs a licensed COMSOL with the RF Module.
  - Then `python sim/cosim.py <case> --solver comsol` chooses the capacitors exactly as for openEMS.
  - `--dry-run` checks the files and writes the planned COMSOL calls without COMSOL.
- **Written without access to COMSOL.** The COMSOL API names in the build script are marked UNVERIFIED. Check the first build in the COMSOL GUI, then run check V25 (COMSOL vs openEMS).

New in v1.9 (crossover bumps, 2026-10-21):

- Where two coils cross, the higher-numbered coil now hops outward over the other, by wire + insulation plus a 10% margin, over a short smooth bump. Coupling and B1 are computed on the bumped wires.
- Before v1.9 the wires of overlapping coils could pass through each other. This was found when a full-wave pair simulation shorted the two coils (DECISIONS.md).
- **Known limitation:** this works for pairs, but in dense knot arrays bumps stack up and some crossings cannot be separated (check V24 fails on arrays). The app then flags the array **"NOT BUILDABLE AS DRAWN"**.

New in v1.8 (knot thickness, 2026-10-21):

- A "Total thickness T" slider lets the knot rise out of its plane like a 3D torus knot, with a side view. Extra height goes outward only.
- T = 0 is identical to earlier versions.
- Self-inductance is now correct for non-flat wires (V23).

New in v1.7 (full-wave capacitor co-simulation):

- Python scripts run openEMS (FDTD) with ports in every capacitor break and a three-layer head phantom.
- A circuit co-simulation then chooses tuning capacitors and a matching network with preamp decoupling.
- It reports how the capacitors change coupling (S21, mutual inductance) and B1, compared with the quasi-static model.
- Workflow: DEVELOPER_GUIDE “v1.7”.
- Every result is **exploratory**: one solver (plus a mariepy cross-check), placeholder phantom and component values, no bench data.
- The current validation verdicts are shown in the app banner and in `validation/validation_report.html`. Failures are listed there, including V15 (raw FDTD loop resonance −10%, explained by the thin-wire effective radius and handled by the V15b correction).

New in v1.6:

- **Head and face zones:** a head model with editable eye and mouth exclusion zones. No wire is allowed inside them.
- **Symmetric rings:** a ring layout mode, mirror-symmetric left–right, with per-ring size, bloom and sharpness.
- **Multi-start solving.**
- **Capacitor breaks:** placed away from crossings, with first-pass values at 1.5, 3 or 7 T, and port export.
- **Save and load** designs.
- **B1:** exact-segment field, cross-checked against Magpylib (V14).

Solver options for the next step (simulation) are in `docs/SOLVER_OPTIONS.md`.

B1 quick look (Array tab): shows the quasi-static Biot–Savart field of the solved wires.

- **Views:** on a placeholder head surface, or as an axial, coronal or sagittal slice.
- **Elements:** combined as root-sum-of-squares, or a single element (click its wire).
- **Use it for:** comparing coverage and depth fall-off between designs. It is not an absolute sensitivity prediction, because tissue and wave effects are not modelled.
- **E-fields are not shown:** this model can’t produce them reliably. Use a full-wave simulator with the exported wire paths.

Solver comparison (pre-registered): `validation/compare_worstpair.html`. The worst-pair term is on by default. It lowers the worst pair but raises the typical (median) coupling; set “Worst-pair weight” to 0 to turn it off.

References

- **Rosa, E. B. & Grover, F. W. (1912).** Formulas and tables for the calculation of mutual and self-inductance (revised). *Bulletin of the Bureau of Standards* 8(1):1–237, Scientific Paper 169. https://nvlpubs.nist.gov/nistpubs/bulletin/08/nbsbulletinv8n1p1_A2b.pdf
  - p. 6, eq. [1]: Maxwell’s coaxial formula, modulus convention. **VERIFIED**.
  - p. 111, eq. [62]: ring self-inductance with surface current. **VERIFIED**.

    - pp. 20–21: Example 1, used for post-hoc check PH1. **VERIFIED**.
- **Roemer, P. B., Edelstein, W. A., Hayes, C. E., Souza, S. P. & Mueller, O. M. (1990).** The NMR phased array. *Magn Reson Med* 16:192–225. doi:10.1002/mrm.1910160203. **UNVERIFIED** (not read).
- **Roemer, P. B. & Edelstein, W. A., US Patent 4,825,162 (1989).** Read via Google Patents OCR text: circular-coil spacing “D approximately equal to 0.75d”. **PARAPHRASE-ONLY.**

Provenance and AI-use statement

This software was written with **Claude Code** (Anthropic’s AI coding assistant, model Claude Opus 5.5) under the direction of **Reza Farivar** (McGill University), who is accountable for it.

- Every source file carries a provenance header.
- Reference formulas were verified against the primary text listed above.
- Validation criteria were pre-registered in `Design.md` (git 39a4e83) before any code was written.
- Failures are recorded in `DECISIONS.md` and in the git history.
- Any use of results in publications must disclose AI assistance as the venue requires (COPE/ICMJE for journals; current Tri-Agency guidance for grants).

Third-party code: Three.js 0.149.0 (MIT) is bundled. Used by scripts but not bundled: openEMS 0.37.0 (GPL-3.0) / CSXCAD (LGPL-3.0), mariepy (GPL-3.0+), Magpylib 6.0.0 (BSD-2), numpy/scipy/h5py/matplotlib, PyTorch. See NOTICE.txt.